SUMMARY · MK 5 DECISION MAKING & NEGOTIATION · WEEK 1

# Introduction to Decision Making: Heuristics & Biases

Why study decision making: people use heuristics, fall into biases and have bounded rationality. Class games, intuition, the three Tversky–Kahneman heuristics, Gigerenzer on risk vs uncertainty, definitions, decision levels, and descriptive/normative/prescriptive approaches.

## Material

### 1 What the class games show

Games show two thinking systems, anchoring (both products = 40,320), persistent illusions, intuitive errors (5 machines → 5 minutes), framing (both Hitler dates false) and priming (S__P → SOUP).

### 2 Two thinking systems at a glance

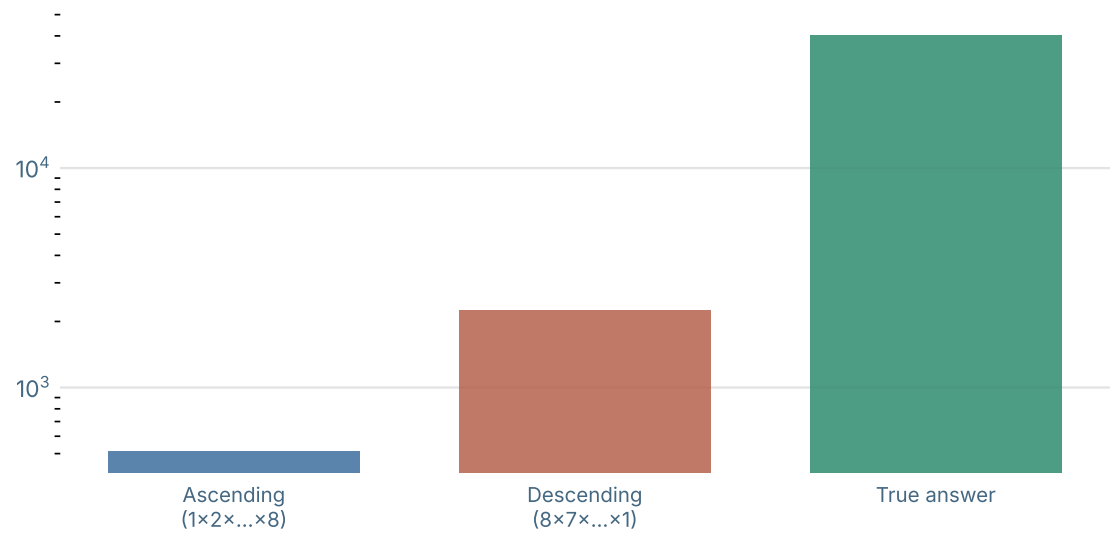
Deep dive

**System 1 — fast**
Automatic, effortless, parallel, emotional.

vs

**System 2 — slow**
Deliberate, effortful, serial, limited.

**Anchoring: median guesses vs the true answer**




Tversky & Kahneman (1974). Log scale.

Median guesses in Tversky & Kahneman (1974): 2,250 (descending) vs 512 (ascending); the true answer is 40,320. Adjustment from an anchor is insufficient.

## 3 Intuition and heuristics

Intuition is subconscious, parallel, emotional pattern matching; useful under time pressure, change, poor structure, ambiguity or no precedent. Heuristics: availability, representativeness, anchoring & adjustment — useful shortcuts that bias judgement when careful statistical reasoning is needed. Gigerenzer: under risk use models; under uncertainty robust heuristics often win.

## 4 Heuristic → bias → antidote Deep dive

| Heuristic | Bias | Antidote |
|---|---|---|
| Availability | Over-weighting vivid events | Look at base rates |
| Representativeness | Base-rate and sample-size neglect | Ask about proportions and n |

| | | |
|---|---|---|
| Anchoring | Insufficient adjustment | Estimate first; use a counter-anchor |

**Linda**
"Bank teller and feminist" cannot be more likely than "bank teller" — the conjunction fallacy.

↕ Top row: stable world · bottom: fast-changing

**Risk**
Statistics and optimisation.

**Uncertainty (stable)**
Simple heuristics work.

**Risk (changing)**
Recalibrate models.

**Uncertainty**
Fast-and-frugal heuristics can win.

↔ Left column: probabilities known · right: unknown

## 5 What decision making is, and its levels

Definitions from Simon, Tryfos, Wang & Ruhe. Strategic decisions are broad, complex, enduring, delayed, contested and political. Descriptive (what people do), normative (what they should do in theory), prescriptive (what they should and can do — the course focus). Rational decision making weighs criteria and picks the highest utility.

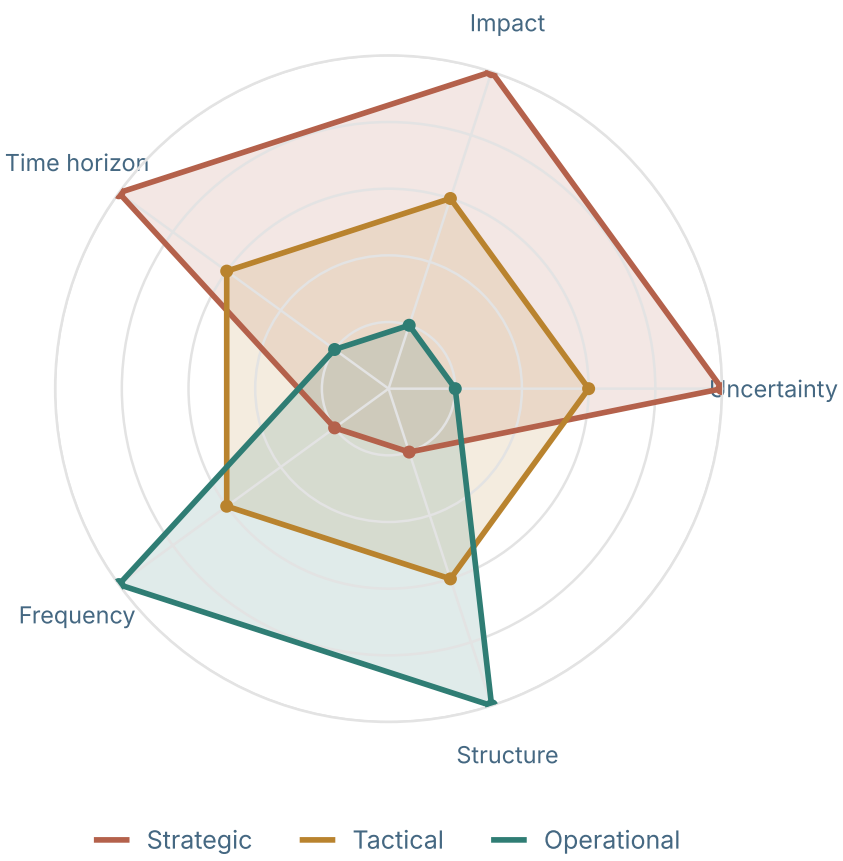
## 6 Decision levels and approaches to studying decisions Deep dive

**Strategic** — top management, long-term, unstructured

**Tactical** — middle managers

**Operational** — supervisors, routine

**Decision traits by level (conceptual, 1–5)**




**Descriptive**
How people decide.

**Normative**
How rational agents should decide.

**Prescriptive**
Practical tools to help real people decide better.

## Key takeaways

1. Bounded rationality, heuristics, biases.
2. Availability, representativeness, anchoring.
3. Risk ≠ uncertainty.
4. Descriptive, normative, prescriptive.
5. Strategic decisions are broad and irreversible.

## Case studies

Case & worked answers

### Reflection: easiest vs hardest decision

Easy decisions are repetitive, low-stakes, reversible; hard ones are one-off, multi-stakeholder, uncertain and irreversible — the same traits as strategic decisions, which call for prescriptive tools. Ask which heuristic shaped your own hard decision.

## Practice questions

Try answering first. The answer key is below each question.

**1. Judging plane-crash risk from news coverage is...**

A. Anchoring
B. Availability
C. Representativeness
D. Satisficing

**Answer: B.** Easy to recall ≠ frequent.

**2. 5 machines, 5 min, 5 widgets → 100 machines, 100 widgets takes...**

A. 100 min
B. 20 min
C. 5 min
D. 1 min

**Answer: C.** Each machine makes one widget per 5 minutes.

**3. The Linda problem shows which heuristic?**

A. Availability
B. Representativeness
C. Anchoring
D. Affect

**Answer: B.** Similarity beats probability logic.

**4. Per Gigerenzer, simple heuristics shine under...**

A. Risk
B. Uncertainty
C. Certainty
D. Never

**Answer: B.** Less is more under uncertainty.

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**5. The approach offering practical tools for real people is...**

A. Descriptive
B. Normative
C. Prescriptive
D. Intuitive

**Answer: C.** Prescriptive bridges the two.

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**Sources**


- [Material] Week 1 - Intro to Decision Making (Dr. Valid Hasyimi)
- Bazerman, Judgment in Managerial Decision Making, Bab 1–2
- Gigerenzer, Simple Heuristics That Make Us Smart